

MILESTONE 2

Comprehensive Performance & Deliverables Report

AI-Powered Wildlife Population Intelligence System

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Program	Infosys Springboard Virtual Internship Program
Batch	Batch 2
Project	AI Wildlife Population Intelligence System
Repository	github.com/Jayasrip08/wildlife-population-intelligence-system
Milestone	Milestone 2 — Completed

Executive Summary

Milestone 2 of the AI-Powered Wildlife Population Intelligence System delivers the core intelligence layer of the platform: an image-based species recognition engine, a bioacoustic vocalization analysis engine, a biodiversity analytics module, and an automated PDF reporting pipeline. All modules have been implemented end-to-end across the backend and frontend and verified through local integration testing.

This document consolidates the modules built, the API surface exposed, the underlying algorithms and scoring models used, the database schema extensions, and the verification results completed during Milestone 2 into a single reference file, following the same reporting structure established in Milestone 1.

1. Project Overview & SDLC Progress

Building on the Iterative & Agile SDLC model adopted in Milestone 1, Milestone 2 focused on the AI/ML intelligence layer of the system — moving from foundational architecture and authentication into active species detection, acoustic classification, and ecological scoring.

Milestone Roadmap

- **Milestone 1 (Completed):** Project Initialization, Requirement Analysis, Architecture & Database Design, JWT Authentication & RBAC, Dataset Ingestion Pipeline Setup.
- **Milestone 2 (Completed):** Species Recognition Engine (Vision + Audio), Biodiversity Analytics Engine, PDF Audit Report Generation, Frontend & Backend Integration.
- **Milestone 3 (Planned):** Production Model Fine-tuning, High-throughput Video Stream Processing, Edge Device Deployment.

2. System Architecture — Intelligence Layer Additions

Milestone 2 extends the Milestone 1 architecture (React + Vite frontend, FastAPI backend, PostgreSQL/PostGIS + MongoDB data layer) with a dedicated machine-learning and analytics layer sitting behind the existing REST API.

Layer	Component / Stack
Frontend Interface	React + Vite — Species Analysis, Audio Lab, Biodiversity Dashboard, Reports Archive tabs
API Gateway	REST API (JSON / JWT) — FastAPI routers for species, bioacoustics, biodiversity, reports
Vision Engine	PyTorch image pipeline — backend/ml_image.py
Bioacoustics Engine	Librosa audio feature extraction & classifier — backend/ml_audio.py
Analytics Engine	Shannon–Wiener / Simpson diversity & ecosystem scoring — backend/biodiversity.py
Reporting Engine	ReportLab PDF generation — backend/biodiversity.py & main.py
Persistence	PostgreSQL (ORM tables) — image_detections, audio_detections, biodiversity_metrics, wildlife_reports

Component Breakdown

- Wildlife Image Analysis & Species Classification Engine: normalizes camera-trap and drone imagery, generates bounding boxes, counts individuals, and scores image sharpness and behavior (Grazing, Alert, Resting).
- Bioacoustic Recognition & Animal Call Identification: extracts spectral centroids, clip duration, and frequency spectrums, then classifies mammal roars, bird songs, amphibian calls, and infrasonic elephant rumbles with noise filtering.
- Biodiversity Intelligence & Assessment Engine: computes Shannon–Wiener and Simpson diversity indices and rolls them into a weighted, multi-criteria Ecosystem Health Score.
- Wildlife Monitoring PDF Report Generator: assembles executive audit documents with styled tables, benchmark comparisons, and conservation recommendations, served for secure static download.
- Interactive Frontend Workflows: drag-and-drop image upload with live detection feedback, an audio lab with frequency spectrum and match confidence display, live Shannon/Simpson index cards and health progress bars, and one-click report generation and download.

3. Species Recognition & Biodiversity Workflow

Milestone 2 implements the AI Detection and Intelligence stages of the end-to-end wildlife monitoring workflow introduced in Milestone 1:

Stage	Description
1. Media Capture	Image or audio file submitted from camera trap, drone, or bioacoustic node.
2. Vision / Audio Inference	PyTorch vision pipeline or Librosa feature extraction runs against the captured media.
3. Detection & Classification	Species identification, bounding boxes, animal counts, or vocalization category with confidence scores.
4. Biodiversity Aggregation	Shannon–Wiener and Simpson indices, species richness, and ecosystem health score computed across detections.
5. Reporting & Dashboard Sync	Results persisted to PostgreSQL, surfaced on the live dashboard, and available for on-demand PDF export.

4. Requirements Matrix & Specifications

A. Functional Requirements (FR)

ID	Category	Description	Status
FR-06	Image Species Analysis	Detect, classify, count, and quality-score animals in uploaded images via POST /api/v1/species/analyze-image.	 Completed
FR-07	Bioacoustic Analysis	Extract acoustic features and classify vocalizations via POST /api/v1/bioacoustics/analyze-audio.	 Completed
FR-08	Biodiversity Analytics	Compute Shannon–Wiener, Simpson, and Ecosystem Health Score via GET /api/v1/biodiversity/analytics.	 Completed
FR-09	PDF Report Generation	Generate and download executive audit PDFs via POST /api/v1/reports/generate and GET /api/v1/reports/download/{pdf_filename}.	 Completed
FR-10	Frontend Intelligence Dashboards	Species Analysis, Audio Lab, Biodiversity Dashboard, and Reports Archive tabs with live feedback.	 Completed

B. Non-Functional Requirements (NFR)

ID	Attribute	Target Metric / Specification	Verification
NFR-06	Accuracy	Species and vocalization classification returns a calibrated confidence score per detection.	Verified
NFR-07	Performance	Image and audio analysis endpoints respond within an acceptable interactive latency for single-file uploads.	Pass
NFR-08	Data Integrity	All detections and computed indices persist to normalized PostgreSQL ORM tables.	Verified
NFR-09	Reliability	PDF generation and download served securely and deterministically from the FastAPI static layer.	Verified
NFR-10	Usability	Real-time detection feedback and live index cards on the dashboard without manual refresh.	Verified

5. User Stories & Acceptance Criteria

User Story 1: Wildlife Researcher — Image-Based Species Identification

- As a Wildlife Researcher
- I want to upload camera-trap or drone imagery and receive automated species detections with bounding boxes and counts
- So that I can quantify population presence without manual image tagging.

Acceptance Criteria

- Detection results display species tag, confidence score, bounding box coordinates, and animal count.
- Image quality (sharpness) and animal behavior (Grazing, Alert, Resting) are surfaced alongside each detection.

User Story 2: Field Ecologist — Bioacoustic Vocalization Analysis

- As a Field Ecologist
- I want to upload audio recordings and have them automatically classified by vocalization type
- So that I can track vocal species activity, including infrasonic elephant rumbles, without manual spectrogram review.

Acceptance Criteria

- Audio Lab displays spectral centroid, clip duration, and frequency spectrum for each upload.
- Classifier returns a vocalization category (mammal roar, bird song, amphibian call, infrasonic rumble) with match confidence.

User Story 3: Conservation Manager — Biodiversity Reporting

- As a Conservation Manager
- I want a single dashboard view of habitat diversity and an exportable audit report
- So that I can brief stakeholders and regulators on ecosystem health with defensible metrics.

Acceptance Criteria

- Dashboard shows live Shannon–Wiener and Simpson index cards plus an overall Ecosystem Health progress bar.
- One-click report generation produces a downloadable PDF with benchmarks and conservation recommendations.

6. Database Schema Extensions (PostgreSQL ORM)

Milestone 2 introduces four new relational tables to persist detection results, aggregated biodiversity metrics, and generated reports, extending the PostgreSQL schema defined in Milestone 1.

1. Table: image_detections

```
CREATE TABLE image_detections (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  site_id UUID REFERENCES monitoring_sites(id) ON DELETE CASCADE,  
  image_url VARCHAR(500) NOT NULL,  
  species VARCHAR(150),  
  confidence NUMERIC(5,3),  
  bounding_box JSONB,          -- [x_min, y_min, x_max, y_max]  
  animal_count INTEGER DEFAULT 1,  
  sharpness_score NUMERIC(5,2),  
  behavior VARCHAR(30) CHECK (behavior IN ('Grazing', 'Alert',  
'Resting')),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP  
);
```

2. Table: audio_detections

```
CREATE TABLE audio_detections (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  site_id UUID REFERENCES monitoring_sites(id) ON DELETE CASCADE,  
  audio_url VARCHAR(500) NOT NULL,  
  vocalization_type VARCHAR(50),    -- Mammal Roar | Bird Song |  
  Amphibian Call | Infrasonic Rumble  
  species VARCHAR(150),  
  confidence NUMERIC(5,3),  
  spectral_centroid NUMERIC(10,3),  
  duration_seconds NUMERIC(8,2),  
  frequency_band VARCHAR(30),
```

```
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

3. Table: biodiversity_metrics

```
CREATE TABLE biodiversity_metrics (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  survey_id UUID REFERENCES surveys(id) ON DELETE CASCADE,
  shannon_index NUMERIC(6,4),      -- H' = -sum(p_i * ln(p_i))
  simpson_index NUMERIC(6,4),      -- 1 / D
  species_richness INTEGER,
  habitat_quality_score NUMERIC(5,2),
  stability_score NUMERIC(5,2),
  ecosystem_health_score NUMERIC(5,2), -- 30% diversity + 30% habitat
+ 20% richness + 20% stability
  computed_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

4. Table: wildlife_reports

```
CREATE TABLE wildlife_reports (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  survey_id UUID REFERENCES surveys(id) ON DELETE CASCADE,
  generated_by UUID REFERENCES users(id),
  pdf_filename VARCHAR(255) NOT NULL,
  pdf_url VARCHAR(500) NOT NULL,
  summary_notes TEXT,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

7. Milestone 2 Benchmarks & Verification Summary

Comprehensive Deliverable Verification

Task Component /	Target Deliverable	Execution & Result	Status
Image Analysis Engine	PyTorch vision pipeline with detection, counting, and behavior tagging	Implemented backend/ml_image.py with bounding-box, count, sharpness, and behavior scoring	PASS
Bioacoustic Engine	Librosa feature extraction & vocalization classifier	Implemented backend/ml_audio.py with	PASS

Task Component /	Target Deliverable	Execution & Result	Status
		spectral centroid, duration, and frequency-band analysis	
Biodiversity Analytics	Shannon / Simpson indices & weighted health score	Implemented backend/biodiversity.py with multi-criteria ecosystem scoring model	PASS
PDF Reporting	Automated executive audit report generation & download	Implemented ReportLab pipeline with styled tables and secure static streaming	PASS
Frontend Integration	Species Analysis, Audio Lab, Biodiversity Dashboard, Reports Archive tabs	Delivered in frontend/src/App.jsx with real-time feedback	PASS
Database Integration	ORM tables for detections, metrics, and reports	Created image_detections, audio_detections, biodiversity_metrics, wildlife_reports	PASS

Backend API Response Verification

Endpoint	Description	Result
GET /health	System telemetry check	{"status": "ok", "milestone": 2}
POST /api/v1/species/analyze-image	Image species detection	Verified
GET /api/v1/species/image-detections	Retrieve stored image detections	Verified
POST /api/v1/bioacoustics/analyze-audio	Audio vocalization analysis	Verified
GET /api/v1/bioacoustics/audio-detections	Retrieve stored audio detections	Verified
GET /api/v1/biodiversity/analytics	Diversity index & health score computation	Verified
POST /api/v1/reports/generate	PDF report generation	Verified
GET /api/v1/reports/download/{pdf_filename}	PDF report download	Verified

Backend verified on FastAPI Uvicorn server at <http://127.0.0.1:8000>; frontend verified on Vite dev server at <http://localhost:5173>, with a successful production Vite build.

8. Environment Setup Reference (.env) — Milestone 2 Additions

The following environment variables extend the Milestone 1 configuration template with settings required for the ML inference and reporting pipelines.

```
# Machine Learning Model Settings
IMAGE_MODEL_PATH=./models/species_vision_model.pt
AUDIO_MODEL_PATH=./models/bioacoustic_classifier.pkl
MODEL_DEVICE=cpu

# Media Storage
IMAGE_UPLOAD_PATH=./uploads/images
AUDIO_UPLOAD_PATH=./uploads/audio

# Reporting
REPORTS_STORAGE_PATH=./reports
REPORTLAB_PAGE_SIZE=A4
```

Note: model weight files and generated report artifacts are git-ignored in the submission copy; only path configuration is shown here.

Conclusion & Transition to Milestone 3

Milestone 2 successfully delivered the full AI intelligence layer of the Wildlife Population Intelligence System — image-based species recognition, bioacoustic vocalization classification, biodiversity analytics, and automated PDF reporting — fully integrated with the Milestone 1 foundation and verified end-to-end across backend, frontend, and database layers.

This solid intelligence layer enables the system to proceed to Milestone 3: Population Intelligence, Density Mapping & Conservation Recommendations, which will build on the detection and analytics pipelines established here.